

Remote Sensing of the Environment

Altimetry Techniques & Analysis Lab

During this lab, we will look at several different altimetry techniques for measuring surface elevations on a planet:

- Using photoclinometry (shape-from-shading) to estimate surface elevations
- Looking at laser altimetry data from the MOLA instrument.
- Using stereo topographic data derived from MOC stereo images
- Deducing some first-order properties about geomorphic features on Mars.

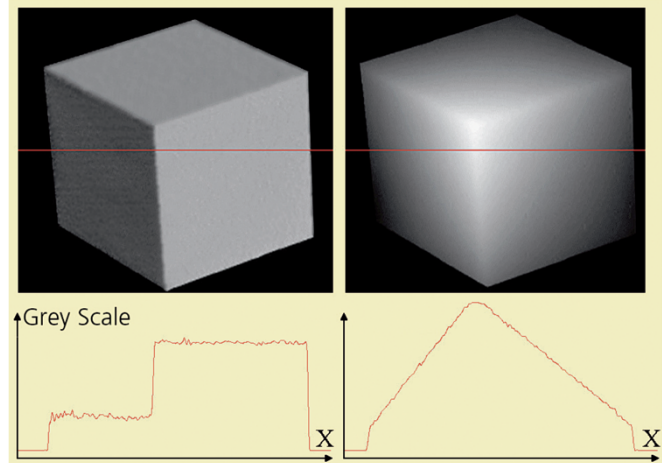
I - Geologic Background and Imaging Data

In this lab, we will look at some of the different techniques for measuring elevation via remote sensing. We'll be working in a location in eastern Arabia Terra, Mars, within the Tikhonravov impact crater. To get an overview of the geological setting, look at the image m0806186.jpg in the lab directory. This is an image taken in the Arabia Terra region of Mars in October 1999 by the Wide Angle Mars Orbiter Camera (MOC), which flew on the Mars Global Surveyor (MGS) mission. (It is a common setup on spacecraft to have a wide angular field-of-view, but low angular resolution, camera paired with a narrow angle, high resolution camera.)

Now, take a look at the corresponding Narrow Angle image, M0806185.gif. **Q1: Describe the what geological features you observe in the Narrow Angle image** (we understand this is not a geology class! But we do hope to sharpen your image interpretation skills). The Wide Angle camera was a standard framing camera, much like your cell phone or personal camera. The companion narrow angle MOC instrument was a pushbroom camera - it has one row of detectors that sweeps across the surface of the planet as the spacecraft moves in its orbit. The ground resolution along this line of detectors (known as crosstrack resolution) is set just by the height of the orbit - if the spacecraft is far away from the planet each pixel sees more of the surface. The mapping orbit was planned to be around 400 km altitude, however there is ~30 Km of relief on the planet (from the summit of Olympus Mons to the bottom of the Hellas impact basin) so the crosstrack resolution can vary from place to place on the planet (by almost 10%). The downtrack resolution is determined by the groundspeed of the spacecraft ($\sim 3 \text{ km s}^{-1}$) combined with the length of time the camera exposes each line. Ideally, you want the spacecraft to move only a fraction of a pixel (to prevent smearing) in the downtrack direction during an exposure, however the exposure must be long enough to collect sufficient light for a high signal-to-noise ratio.

II – Photoclinometry

This section is aimed at being an introduction to the concept of 'photoclinometry' (deriving topography from surface shading). The image on the left give a sense of how the brightness in an image (left, with a transect shown in the graph), can be used to infer 3D shape (right), when illuminated from a single source. Photoclinometry is difficult to do in an accurate way, but it has the advantage that it can be done with just a single image, and it can be used to infer topography all the way down to the single-pixel scale. This exercise will illustrate some of the qualitative concepts associated with it. After navigating to the location where you unzipped the lab files within Matlab, you need to add the path of the Matlab routines for this lab by typing:



`addpath matlab`

There is a Matlab program called `pclin.m` among the lab files. This is a basic photoclinometry program which will allow you to vary some input parameters and see their effects. This program uses the surface reflectance (I/F) values from the image `m0806185.lev1.cub`. Some of the program lines are reproduced below:

```
res = 2.95; %Size of a pixel in meters
sunaz = 320.65; % Solar Azimuth
inc_ang = 48.18; % Inclination Angle
```

These lines set up the variables specific to this image. The solar azimuth is important, as only profiles that cut across the image parallel to the illumination direction can be used for deriving topography from photoclinometry. **Q2: Explain briefly why this is the case.**

```
b = b-shadow;
```

`b` is the variable that stores the I/F values along the sun's line of sight. Here we subtract some portion of the I/F value ("shadow"), which is due to scattering by the atmosphere (called shadow brightness because this is the only way shadows are illuminated at all).

```
z = acos(b.*pi.*cos(inc_ang*(pi/180.0))./albedo);
z = z - (inc_ang*(pi/180.0));
z = res.*tan(z)
for i=2:length(z)
    z(i) = z(i-1)-z(i);
end
```

These few lines are the guts of the program. The first calculates the incidence angle for each pixel based on its I/F value. The second then removes the incidence angle that a flat surface would have, leaving just the local slopes for each pixel. The third line converts the slope of each pixel to the change in height of that pixel. The for loop adds up all these height changes to find the actual height of each pixel. This isn't a programming class, but we can think about the factors that are important for the technique. Notice for example that we have to assume some value for the albedo and that the atmosphere can have an important role, due to its ability to scatter light. You can start the program by typing:

```
pclin('m0806185.lev1.cub')
```

at the matlab prompt. The default case will come up - the program selects a default part of the image, guesses an albedo and assumes no atmospheric effects. You can force it to use some particular albedo by calling it like *pclin('m0806185.lev1.cub',0.25)* for an albedo of 25%. You can use *pclin('m0806185.lev1.cub',.25, 3000)* to select a starting line for the section of image you want to look at (line 3000, in this case). You can also make some correction for the atmosphere by assuming some amount of the image brightness is due to scattering, e.g.

pclin('m0806185.lev1.cub',.25, 3000,.02) will assume that 0.02 needs to be subtracted from each I/F value as a first order atmospheric correction. You can choose each of these three variables, although they must be in the order: albedo, starting line, shadow brightness. (If you want to specify the starting line but not the albedo, for instance, entering -1 will cause it to choose the default albedo value. *pclin('m0806185.lev1.cub',-1, -1, -1)* will cause the program to use the defaults for all three.) Note that the elevation here is in arbitrary units! It's easy to get relative elevations with photoclinometry, but this technique isn't reliable enough to actually measure heights in meters, without reference points. So in discussing these heights, don't assume some unit, it's just an arbitrary scale. Experiment with different albedos, letting the program guess first and then trying higher and lower values. Try this for a number of areas and try to figure out what goes wrong when the albedo is far away from the real answer. **Q3: Explain it as best you can in words.** Do the same for the shadow illumination parameter. Export your relevant MATLAB plots when you're done, using File -> Export on the plot window.

III – Laser Altimetry

The Mars Orbiter Laser Altimeter (MOLA) also operated on the MGS spacecraft. It measured topography by pulsing a laser at the planet's surface and recording the length of time it took for that pulse to be reflected back. It was a nadir-pointing instrument and so could only see what was directly below it. The laser fired 10 shots a second while the spacecraft had a groundspeed of 3 km s^{-1} , so each shot is separated from the next by about 300 meters. The laser illuminated a patch on the surface that's about 120 meters in size. As the spacecraft orbited the planet a track of elevation measurements was built up of the terrain that MGS had passed over. MOC images and topographic data were taken at the same time, which allowed them to be precisely co-registered, which was quite useful for interpreting landforms. We'll look at one of the single laser tracks that crosses over the image from part I. However, this data

is only along the track of the spacecraft (the long axis of the image), and only along a small strip of the image. (The accumulated volume of millions of MOLA shots was eventually gridded into one planet-wide map.) The data structure of the MOLA files is long and complicated so to spare you the details, I have extracted just the relevant altimeter spots, and written a program to plot everything together in Matlab. In Matlab type:

```
[time,y,x,z,d]=mocmolaplot('m0806185');
```

Some plots will pop up, and some arrays will be spit out, including the projected x and y coordinates, the corresponding elevations and times (in J2000, or seconds after 00:00:00 1/1/2000), and along track distance. The plot on the left hand side of the figure simply plots the MOC image we have been looking at, now projected, with the MOLA points on top. The actual footprint of the laser beam is significantly larger than the points shown here, but I think this makes for easier viewing. But if a spot falls right on the edge of a cliff, the results may look a little funny. The plot on the right shows the elevation of the MOLA data vs. Latitude. You can now see the actual elevations of this image. Now plot the elevation on your own; you can use the command:

```
plot(d,z)
```

This will plot the topography recorded against the distance from the first point. (You can open a new figure for plotting data with the convenient command 'figure'.) Remember each point represents about 300m along the ground, or 0.1 seconds. Notice in this dataset that we are dealing with absolute elevations, as opposed to unscaled and relative elevations in the photogrammetry section. This is very useful! **Q4: So, what is the total relief along this track?** Because we now have absolute elevations, we can compare different parts of the image to each other. **Q5: Are all of the high points in this image at the same elevation? What are the maximum slopes seen in this track, and are they consistent with loose sediment (less than ~30 degrees, as opposed to bedrock which can support steeper slopes)?** The track isn't along the direction of maximum slope, so this will only be a lower limit on the actual maximum slopes. Too bad we don't have topographic data for the whole image... The takeaway here for laser altimetry is that it is great for precision, but the coverage and spatial resolution may be limited **Export your relevant MATLAB plots when you're done.**

IV. Stereogrammetry

The third topographic technique we will employ is stereo imagery. It just so happens that the image we have been looking at has been imaged twice by the MOC team. Images from different angles can be combined to extract topographic information, as in a 3-D picture. By measuring the parallax between various features, and with precise knowledge of the camera position, good topographic data can be obtained, at near the resolution of the original MOC images. Advantages to this method are the higher resolution, and the ability to extract absolute elevations. However, the extent of MOC stereo coverage is rather limited, and it does not have quite the vertical precision of the Laser Altimeter data. We will take a look at this data, and you

can see the advantages and disadvantages for yourself. To save you time, I've made a stereo Digital Elevation Model (DEM) beforehand. In Matlab, type:

```
stereoplot
```

This will bring up two images. The first one is our friend M0806185. Again, this is the projected version of the image. The second plot is of the stereo DEM, with the image behind it. This is pretty different than the other two analysis methods! We have full 2-D coverage of the image...sort of. Notice that there are lots of areas where the automatic correlator couldn't find matches between the images. **Q6: Why do you think this is?** You can look at the corresponding image, E0600098, which I've included in your directory to better understand what the computer is trying to match here. Also, you can see there is some noise to the data, much more than the MOLA data. We're going to use a function called *proftool3*, so you can take profiles of this plot in any direction you want. It can be used in two ways. Close the second (color) plot window, or just make the black and white image the active plot. Type:

```
proftool3(z)
```

Use the mouse to select the profile in the image window (right click when you're done to exit). This function lets you click on the image for reference, but extract the corresponding topography (from the elevation matrix, *z*). Follow the directions that come up in the matlab workspace. A plot should pop up showing the elevation profile. Alternatively, you can type:

```
[x_data,y_data,z_data,d_data]=proftool3(z);
```

This usage lets you save the *x*, *y*, *z*, and distance values of the profile in the four output arguments, in case you want to plot them again later. Let's try two analyses of the terrain, using this tool: **Q7: 1) Determine whether the layers exposed along the edge of this cliff are roughly horizontal or not. 2) Determine if the angle of the slope streaks which can be seen throughout the image is this what you would expect for a fine dust layer on the surface (roughly 30 degrees for loose sediment)?** Again, export your plots to support your conclusions for these questions. Finally, type `hist(z(:),20)` to view a histogram of elevations for this DEM. **Q8: What are the maximum and minimum elevations?**

V. The bottom line

This lab is designed to give you a feel for the positives and negatives of various altimetry techniques from satellite remote sensing. To illustrate these points, the data we used today are from older instruments from around the year 2000, which helps accentuate some of the drawbacks of each technique. However, each of these techniques has improved over the last two decades, due to a combination of advanced computer processing algorithms, better satellite position and attitude control, and improved instrumentation. Each of these techniques remain in use, and can now be quite precise by registering the data sets to each

other. Typically in a planetary application, stereo topography will be registered to lower resolution, but more accurate laser altimetry data, and then photoclinometry may be used to refine the very finest-scale topography from the images.

Phew! Congratulations, you're done.

Please let me know if you find errors or problems with the text of the lab, or with the programs.

That's it, good luck!

-Kevin